Minor in Computer Networks

Rationale

In today's world, we cannot imagine a company without computer networks. Therefore, networking skills are becoming a valuable asset for employees across domains.

In response to national, regional, and international needs and demand for experts in computer networks, the College of Engineering/ECE Department and the College of Arts and Sciences/CS Department offer a minor in Computer Networking.

Program's Purpose

The main aim of the minor is to expose students from different backgrounds to the concepts of computer networks and the protocols that allow devices and systems to communicate. It will equip them with the knowledge and skills needed to install and operate networks at various scales. Furthermore, the minor will enable students to earn the Cisco CCNA Routing and Switching certification. The minor shall also serve as good preparation for further and more advanced graduate and/or professional studies in computer networks.

Learning Outcomes

At the end of this minor, the student is expected to demonstrate the ability to:

- Compare different computer network protocols;
- Design simple computer networks;
- Perform simple computer network troubleshooting;
- Investigate simple computer network attacks.

Minor Requirements

This minor is structured to accommodate undergraduate RHU students in the BE or BS program from different disciplines (engineering, computer science, or business). It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements.

Early in their studies, interested RHU students need to declare their intent to pursue a minor in computer networks by completing the relevant minor declaration form and informing their advisor.

To complete the computer networking minor, a student must

1. Declare a minor in computer networks by completing the Minor Declaration Form;
2. Obtain the approval of the Dean of the student's college major and the Dean of the College of Engineering;

3. Earn a Minor Cumulative Grade Point Average (MCGPA) of no less than 70%;
4. Complete 19 credits of coursework as specified below.

It should be noted that common courses between the student's major requirements and the computer networks minor requirements count toward both the major and the minor. The student must complete a minimum of 3 credit hours of coursework that is not counted toward the requirement for their major or any other minor.

Career Options

Computer network experts are well-suited to work in a variety of sectors, including but not limited to telecommunications, hospitals, banks, the police, and the army.

Curriculum and Program

The computer networking minor consists of 5 courses and 4 labs (19 credits), of which 7 credits are mandatory, and the remaining credits are elective, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

Mandatory Courses

The mandatory courses (7 credits) are:

CCEE 354	Computer Networks	3(3,0)
CCEE 354L	Computer Networks Lab	1(0,2)
CCEE 554L	Computer Networks Lab 2	1(0,2)
CCEE 555L	Computer Networks Lab 3	1(0,2)
CCEE 556L	Computer Networks Lab 4	1(0,2)

Elective Courses

Students may choose elective courses (12 credits) from the following list.

CCEE 411	Web Programming	3(3,0)
CCEE 514	Advanced Web Programming	3(3,0)
CCEE 547	Optical Communications	3(3,0)
CCEE 552	Network Security	3(3,0)
CCEE 553	Advanced Communication Networks	3(3,0)
CCEE 555	Advanced Computer Networks	3(3,0)
CCEE 556	Network and System Administration	3(3,0)

Courses Description

CCEE 411	Web Programming	3(3,0)
This course teaches students how to develop and implement web-based programs with an emphasis on interface programming. It introduces students to web development and the client-side and server-side languages and technologies needed to build functional, responsive websites. The course covers HTML5, CSS3, JavaScript/jQuery, PHP, and responsive design. **Co-requisite:** CCEE 214. Equivalent to COSC 333.		

CCEE 354	Computer Networks	3(3,0)
This course enables students to gain fundamental knowledge of computer networks, appreciate the trade-offs and choices in networking, learn to design and analyze protocols, and prepare to study advanced topics in the field. **Co-requisite**: CCEE 221. Equivalent to COSC 360.		

CCEE 354L	Computer Networks Lab	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications. The student will acquire knowledge of network equipment and protocol functionality, learn to build a simple Ethernet network using routers, switches, and computers, and use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts. Accompanying Lab for CCEE 354. **Co-requisite**: CCEE 354.		

CCEE 514	Advanced Web Programming	3(3,0)
This course focuses on server-side programming. It teaches students how to connect their website or web application to a database and save and retrieve data from it. The course exposes students to web controls, validation controls, data source controls, data binding controls, state management, and working with third-party media such as XML and web services. **Prerequisite**: CCEE 411/COSC 333. Equivalent to COSC 434.		

CCEE 547	Optical Communications	3(3,0)
This course is designed to progress from the description of the components in a fiber link to the interconnections within a link or a network. The components of fiber links will be discussed (fiber, connections, sources, and receivers). These optical components will be integrated into a complete optical link. **Prerequisite**: CCEE 331.		

CCEE 552	Network Security	3(3,0)
Network security is the protection of the underlying networking infrastructure from unauthorized access, misuse, or theft. It involves creating a secure infrastructure for devices, applications, and users to operate securely. **Pre-requisite**: CCEE 354 or COSC 360.		

CCEE 553	Advanced Communication Networks	3(3,0)
In this course, we will explore current network architectures and topologies, such as telephone systems (PDH/SDH, SONET), Frame Relay, ATM, and MPLS, as well as technologies under development. We will also learn how to assess network performance and which technologies are used to enhance service quality. At the end of the course, the student will be able to identify the technologies to be used in a given environment and study its performance. **Prerequisite**: CCEE 354.		

CCEE 554L	Computer Networks Lab 2	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications (CCNA 1). The student will acquire knowledge of network equipment and protocol functionality, learn to build a simple Ethernet network using routers, switches, and computers, and use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts through hands-on skills using hardware devices. **Co-requisite**: CCEE 354L.		

CCEE 555	Advanced Computer Networks	1(0,2)
This is an advanced computer networks course that requires active student participation. In this course, we will explore TCP; Internet inter-domain routing; Internet quality of service and traffic engineering; Internet traffic measurement and analysis; data centers; the analysis and performance of content distribution networks; and Software-defined networking. In addition, the students will learn how to build client/server computer network applications using sockets. **Prerequisite**: CCEE 354.		

CCEE 555L	Computer Networks Lab 3	1(0,2)
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 2) Certification. The primary focus of this Lab is routers and switches in small networks. The student will learn the architecture, components, and operations of routers and switches, as well as their basic configuration and functionality. They will also learn how to configure and troubleshoot static and dynamic routing protocols, access-control lists, VLANs, DHCP for IPv4 and IPv6 networks, and NAT operations. **Prerequisite**: CCEE 354L.		

CCEE 556	Network and System Administration	3(3,0)
This course will help students become system administrators and install, configure, maintain, and troubleshoot a Microsoft Windows Server Operating system. It provides them with the knowledge and skills to work with Active Directory technologies in environments including complex network services. This course also provides students with basic knowledge of installing, configuring, and maintaining Linux operating systems. By the end of this course, students will have all the required materials to submit and pass the MCSA: Windows Server certification exam. **Prerequisite**: CCEE 354.		

CCEE 556L	Computer Networks Lab 4	1(0,2)
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 3) Certification. The primary focus of this Lab is routers and switches in large and complex networks. The student will learn how to configure routers and switches with advanced functionalities. They will also learn how to configure and troubleshoot enhanced switching technologies, first-hop redundancy protocols in a switched network, wireless routers and clients, and routers in complex networks. Finally, they will learn how to manage CISCO IOS software licensing and configuration files. **Prerequisite**: CCEE 555L.		